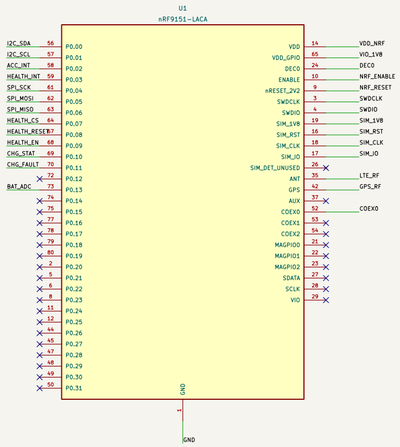
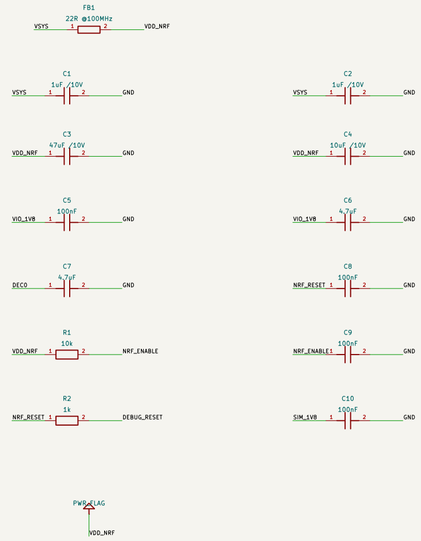
01 / MODEM, GNSS AND APPLICATION MCU

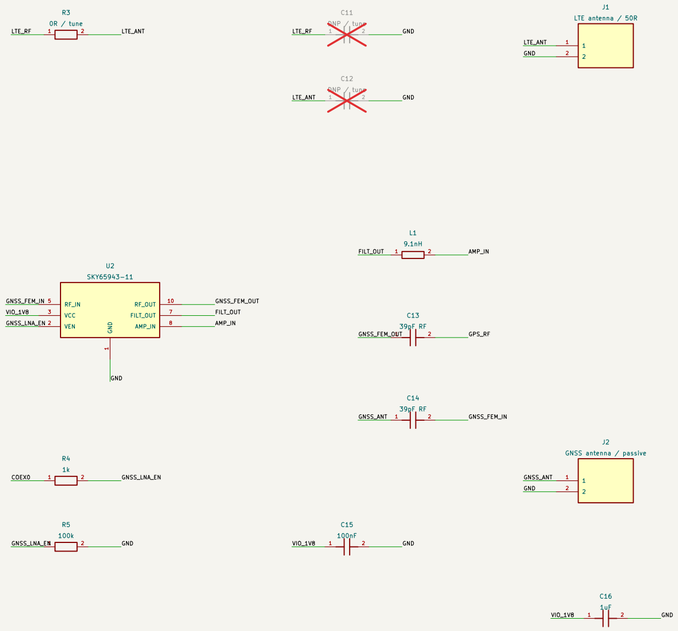


Reserved pins 31–33 and 81–104: electrically unconnected.
SIM_DET is not supported. No external pull-up on nRESET.

REFERENCE DECOUPLING / NOT A COMPLETE RF LAYOUT

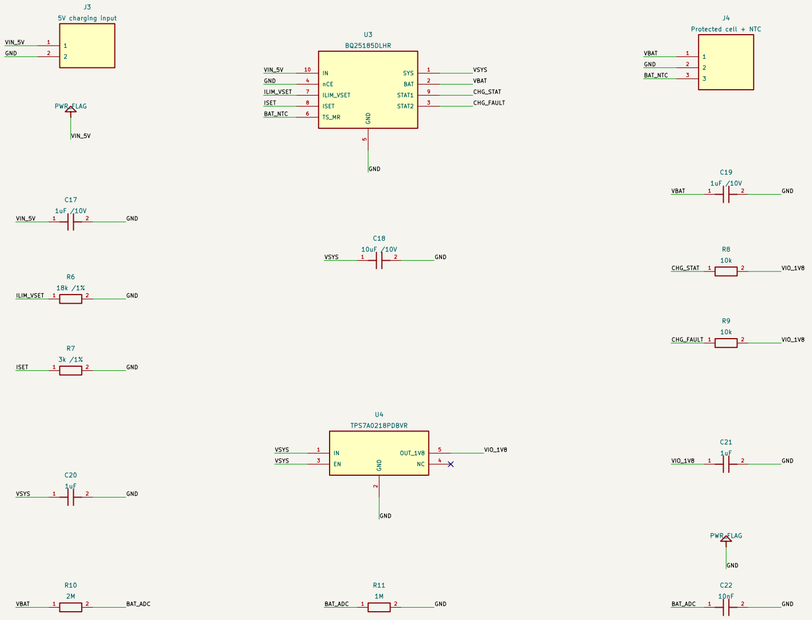


02 / LTE AND GNSS FRONT END



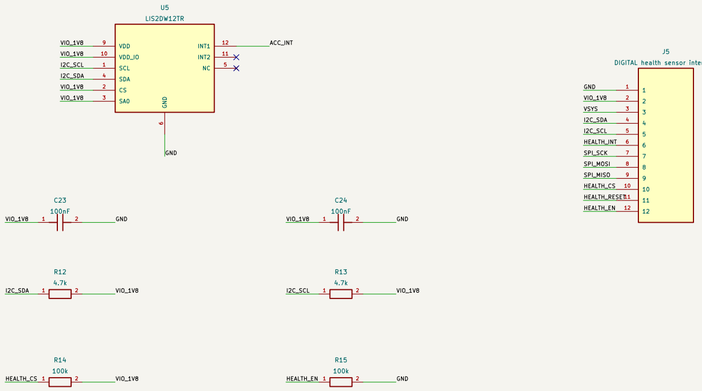
GPS and cellular do not receive/transmit concurrently.
GNSS RF ESD, matching and body/enclosure tuning remain open.
No NTN/B255 requirement in this draft.

03 / POWER — FIRST CIRCUIT DRAFT



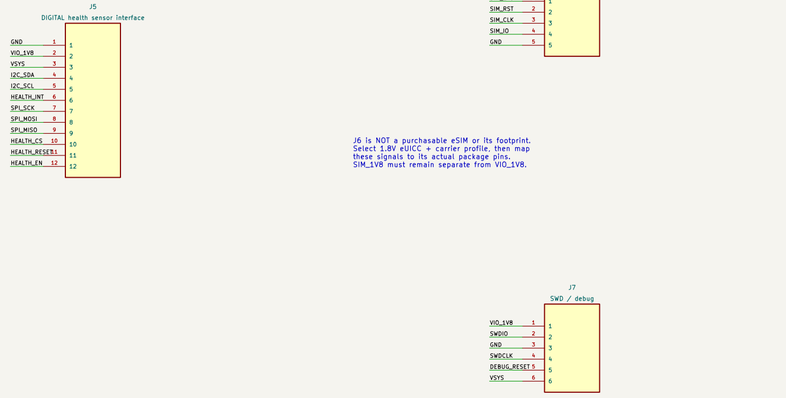
HOLD: add/verify hardware undervoltage cutoff (VDD >=3.0V).
VDD_GPIO startup/shutdown sequencing, charging-input protection,
and battery/NTC ratings before layout release.
BQ25185 internal battery cutoff alone is insufficient for nRF9151.
BAT_ADC: allow divider/filter settling; do not sample immediately.

04 / MOTION AND HEART-RATE EXPANSION



Digital interface only — not an electrode connection.
PPG/ECG measurement head, power switching and protection
must be designed and validated separately.
No heart-rate sensor selected; no heart-rate accuracy claim.

05 / eSIM AND DEBUG INTERFACES



J6 is NOT a purchasable eSIM or its footprint.
Select 1.8V eUICC + carrier profile, then map
these signals to its actual package pins.
SIM_1V8 must remain separate from VIO_1V8.

Reset has an internal 2.2V pull-up.
Debug reset driver: open drain, compatible voltage.
No external reset pull-up.

ENGINEERING DRAFT 0.1 — NOT FOR FABRICATION
Turkey / shared lightweight body / heart-rate feasibility pending

Not for fabrication — unresolved power, eSIM, RF and heart-rate interfaces

Sheet: /
File: collar-tracker.kicad_sch

Title: Collar tracker — schematic review draft

Size: A2	Date: 2026-10-24	**Rev: 0.1-DRAFT**
KiCad E.D.A. 10.0.7		Id: 1/1